Grade 9 Science Assessment: Make an infographic about a planet and its moon(s)

group assessment

Task Overview

Students will design an infographic poster on a chosen planet of our solar system with its moon(s). Each group of 4 students will make one poster.

The poster should contain the following information:

- **planet profile:** Describe the planet's size (diameter), axial tilt, orbit (shape and distance from the Sun), rotation period (day length), revolution period (year length), and planet type (terrestrial, gas giant, or ice giant).
- **Moon profile:** Describe the moon(s): number, size, orbital period, rotation period, and tilt. Explain how the moon(s) move(s) around the planet.
- **Moon–Planet–Sun Interactions:** Explain how the moon(s) and the Sun affect the planet. Consider orbital motion and gravitational force.Explain tides (tidal forces caused by the moon and Sun). Explain how the moon would appear in the sky at different times (phases, apparent size, motion).
- Explain the **impact of tilt, rotation, and revolution** on the planet (e.g., seasons, day/night cycle, year length).
- **Mathematical Calculations:** Calculate the average orbital speed of the planet around the Sun. Calculate the light travel time from the Sun to the planet. Show the correct equations, substitution, and units.
- Include at least **3 fun facts** about the chosen planet or its moon(s). Each fact must be accurate, explained, and cited.
- **Sources:** List all scientific resources used (websites, books, articles).

Students will be marked as a group.

Assessment Rubric

Marking breakdown:

- **55% scientific content**
- **15% Mathematical calculations**
- **20% poster layout and creativity**
- **10% fun facts**

Criterion	What to show	Excellent	Secure	Developing

Planet Characteristics (15 marks)	Accurate, detailed information about the planet's size, tilt, orbit, rotation, revolution, and type.	12-15 Marks Accurate, detailed, and clear information	7-11 marks Mostly accurate with minor gaps	0-6 marks Simple or partly incorrect
Moon Characteristics & Motion (10 marks)	Accurate description of the moon(s): number, size, orbit, rotation, tilt, and how it moves around the planet.	8-10 marks Accurate, detailed, and clear information	4-7 marks Mostly accurate with minor gaps	0-3 marks Simple or partly incorrect
Moon-Planet-Sun Interactions (15 marks)	Clear explanation of tides, how the moon appears in the sky, and the impact of tilt, rotation, and revolution.	12-15 marks Accurate, detailed and clear	7-11 marks Mostly accurate with minor gaps	0-6 marks Simple or partly incorrect
Scientific Vocabulary & Accuracy (5 marks)	Correct use of at least 8 key scientific terms (e.g., orbit, rotation, revolution, tilt, tidal force, eclipse, phase, gravitational force).	5 marks 8+ terms used correctly.	3-4 marks 5–7 terms used correctly.	0-2 marks Fewer than 5 terms used correctly.
Average Orbital Speed Calculation (8 marks)	Correct equation, substitution, working, units, and answer.	7-8 marks Fully correct with clear working.	4-6 marks Mostly correct with minor errors.	0-3 Incorrect or missing work.
Light Travel Time Calculation (7 marks)	Correct equation, substitution, working, units, and answer.	6–7 marks Fully correct with clear working.	3–5 marks Mostly correct with minor errors.	0–2 marks Incorrect or missing working.
Poster Layout & Creativity (15 marks)	Visuals are clear, inviting, and well organized. Information is balanced with visuals. Original student work — no AI-generated content.	12–15 marks Very clear, well organized, original.	7–11 marks Slightly cluttered but still clear. Limited originality.	0–6 marks Poorly organized or AI-generated.
Fun facts (10 marks)	At least 3 accurate, explained, and cited fun facts.	9-10 marks 3+ fun facts, accurate, explained, cited.	6-8 marks 2 fun facts, mostly accurate	0-5 marks 0-1 fun fact or inaccurate
Sources & Referencing (5 marks)	All sources cited correctly (author, title, URL, date accessed).	5 marks All sources cited correctly	3-4 marks Most sources cited.	0-2 marks Few or no sources cited.

Student Instruction Worksheet

Project goal: Create an infographic about a planet in our solar system and its moon(s). The poster must contain accurate information about the planet's size, tilt, orbit, revolution speed, and planet type. It must also contain mathematical calculations for the average orbital speed and light travel time between the Sun and the planet.

Checklist Poster

- The poster has a clear title and shows the planet and its moon(s) clearly.
- The size, tilt, orbit, rotation and revolution speed of the planet are clearly described.
- The moon(s) are described, including how they moves around the planet, theit tilt, orbit time and rotation speed.
- The impact of the moon(s) on tides and how the moon would look in the sky at different times is explained.
- The calculations for the average orbital speed and light travel time are using the correct equations and units.
- At least 3 fun facts are included, explained and cited
- All sources are listed.

Equations You Will Need

Average orbital speed: $v = 2\pi r / T$
v = average orbital speed (m/s)
r = average orbital radius (m)
T = orbital period (s)

Light travel time: $t = d/c$
t = time (s)
d = distance from the Sun to the planet (m)
c = speed of light = 3.0×10^8 m/s

Useful conversions:
1 AU = 1.496 × 10^11 m
1 km = 1000 m
1 day = 86,400 s
1 year = 3.15 × 10^7 s

Success Criteria

□ I can clearly and accurately explain my planet and its moon(s) using an infographi

□ I correctly use at least 8 key scientific terms (e.g., orbit, rotation, revolution, tilt, tidal force, phase, gravitational force).

□ I can explain how the moon affects the planet (tides, sky appearance) in detail.

□ I can calculate the average orbital speed and light travel time for my planet, showing the equation, substitution, and units.

□ My group's infographic is neat, organized, and uses enough visuals that the topic is immediately understandable.

□ I have cited all my sources correctly.

□ My fun facts are accurate, explained, and cited.

Glossary

Orbit	The curved path of an object around a star, planet, or moon.
Rotation	The spinning of an object on its axis.
Revolution	The movement of an object around another object.
Axial tilt	The angle between an object's rotation axis and its orbital plane.
Tidal force	The gravitational pull that causes bulges in a body, leading to tides.
Phase	The apparent shape of the Moon or a planet as seen from Earth.
Gravitational force	The attractive force between objects with mass.
AU (Astronomical Unit)	The average distance from Earth to the Sun (1.496×10^11 m).
Light travel time	The time it takes light to travel from one point to another.

AI Policy

- AI-generated text, images, or calculations are not permitted.
- All visuals must be student-created (hand-drawn or digitally designed by students).
- All information must be researched from cited sources and written in students' own words.

End of Assessment Document